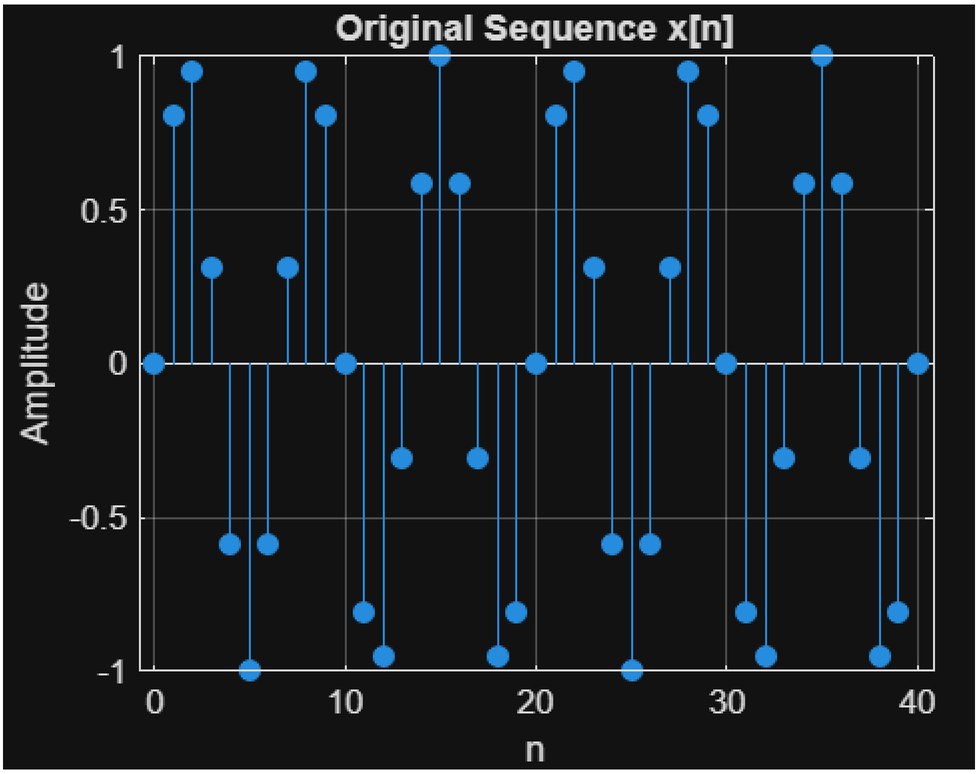
Expt 8	Multirate Signal Processing
Date:	8.1 Implementation and Analysis of Decimation and Interpolation in Multirate Signal Processing Using MATLAB

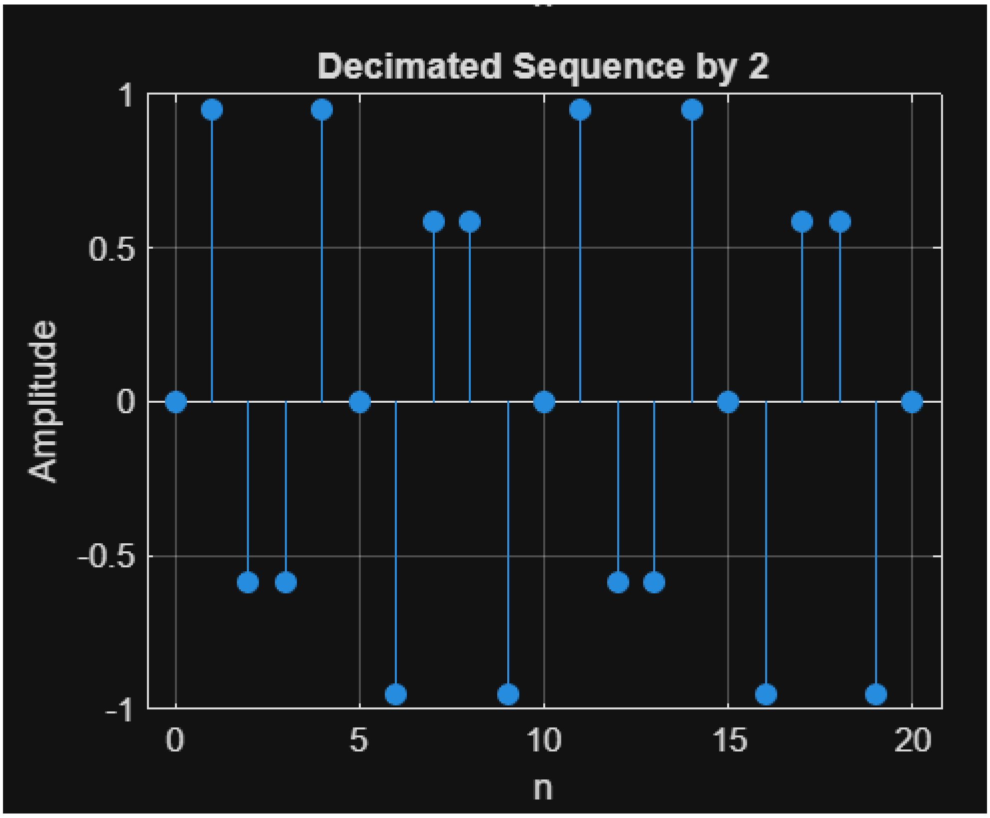
CODE :-

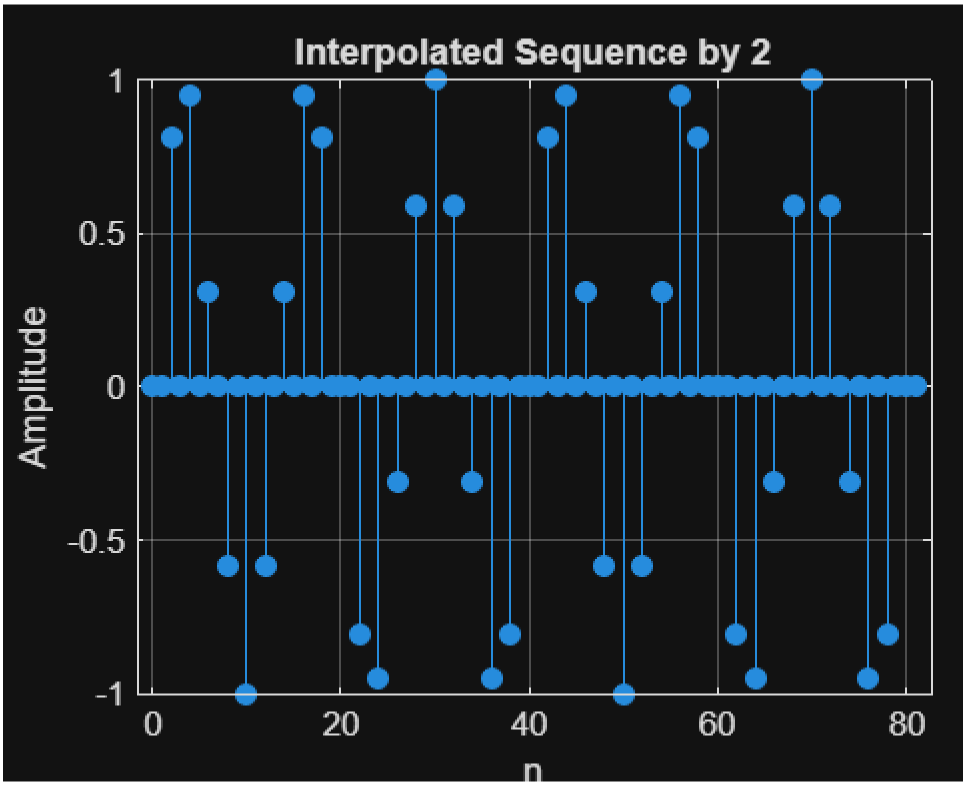
```
clc;
clear;
close all;

% Original Sequence
n = 0:40;
x = sin(0.3*pi*n);

% Decimation by 2
y = x(1:2:end);

nd = 0:length(y)-1;

% Interpolation by 2
v = zeros(1,2*length(x));
v(1:2:end) = x;
ni = 0:length(v)-1;

% Figure 1: Original Sequence
figure;
stem(n,x,'filled');
grid on;
title('Original Sequence x[n]');
xlabel('n');
```

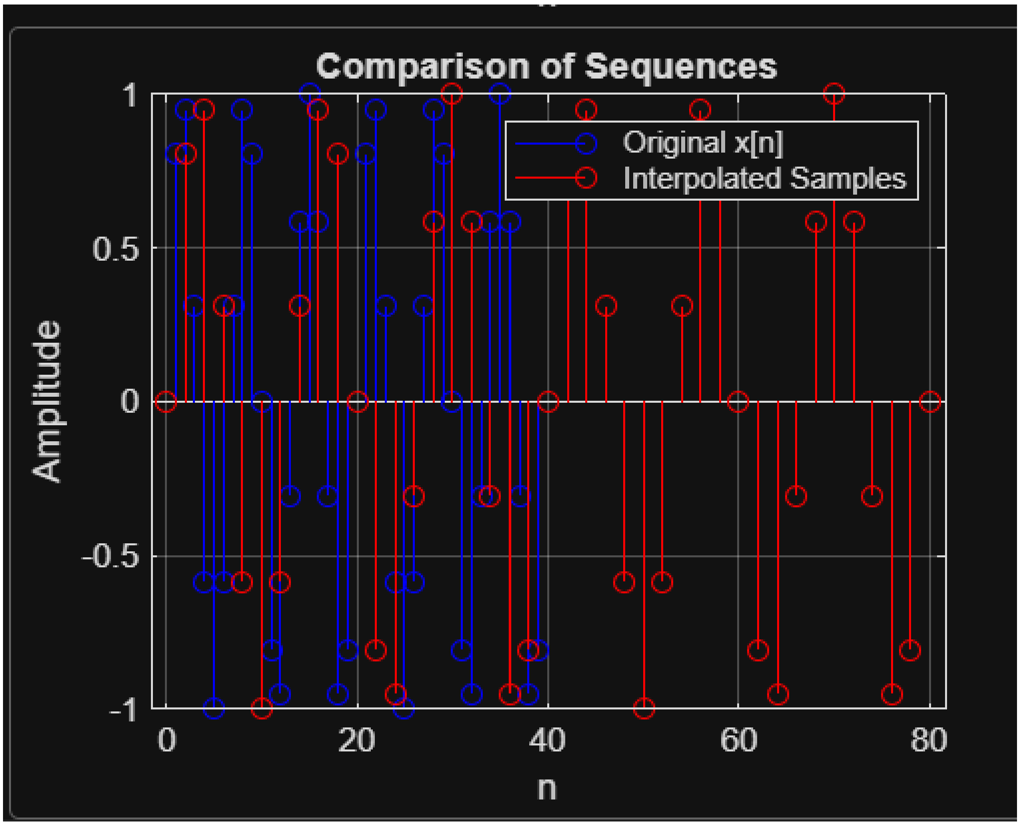
```
ylabel('Amplitude');

% Figure 2: Decimated Sequence
figure;
stem(nd,y,'filled');
grid on;
title('Decimated Sequence by 2');
xlabel('n');
ylabel('Amplitude');

% Figure 3: Interpolated Sequence


figure;
stem(ni,v,'filled');
grid on;
title('Interpolated Sequence by 2');
xlabel('n');
ylabel('Amplitude');

% Figure 4: Comparison
figure;
stem(n,x,'b'); hold on;
stem(0:2:length(v)-2,v(1:2:end),'r');
legend('Original x[n]','Interpolated Samples');
grid on;
title('Comparison of Sequences');
xlabel('n');
ylabel('Amplitude');
```

OUTPUT :-




Decimated Sequence by 2
Amplitude
n
1
0.5
0
-0.5
-1
0
5
10
15
20


Interpolated Sequence by 2
Amplitude
n
1
0.5
0
-0.5
-1
0
20
40
60
80


Comparison of Sequences
Original x[n]
Interpolated Samples
Amplitude
n
1
0.5
0
-0.5
-1
0
20
40
60
80